React.js Installation Guide

Beginner-friendly classroom presentation

Goal: Install React and create your first React application

Recommended setup: Node.js + npm + Vite

By the end: run a React app locally and understand the basic project structure

What Do We Need Before React?

Node.js — JavaScript runtime required by modern React tooling

npm — package manager installed automatically with Node.js

VS Code — recommended code editor

A browser — Chrome, Edge, Firefox, etc.

Internet connection — needed while downloading project dependencies

Step 1 — Install Node.js

Go to the official Node.js website

Download the LTS version for your operating system

Run the installer and keep the default options

Restart the terminal after installation if necessary

```
node -v
npm -v
```

Step 2 — Verify Node.js & npm

Open Command Prompt / PowerShell / VS Code Terminal

Check Node.js version:

Check npm version:

If both commands return version numbers, your setup is ready.

```
node -v
npm -v
```

Step 3 — Create a React App with Vite

Vite is a modern build tool and development server commonly used for React projects.

Open the terminal in the folder where you want to create the project.

```
npm create vite@latest my-react-app
```

Then select: React → JavaScript (or TypeScript if your course uses TS)

Step 4 — Install Dependencies

Move inside the newly created project folder

Install all packages listed in package.json

```
cd my-react-app
npm install
```

Step 5 — Start the Development Server

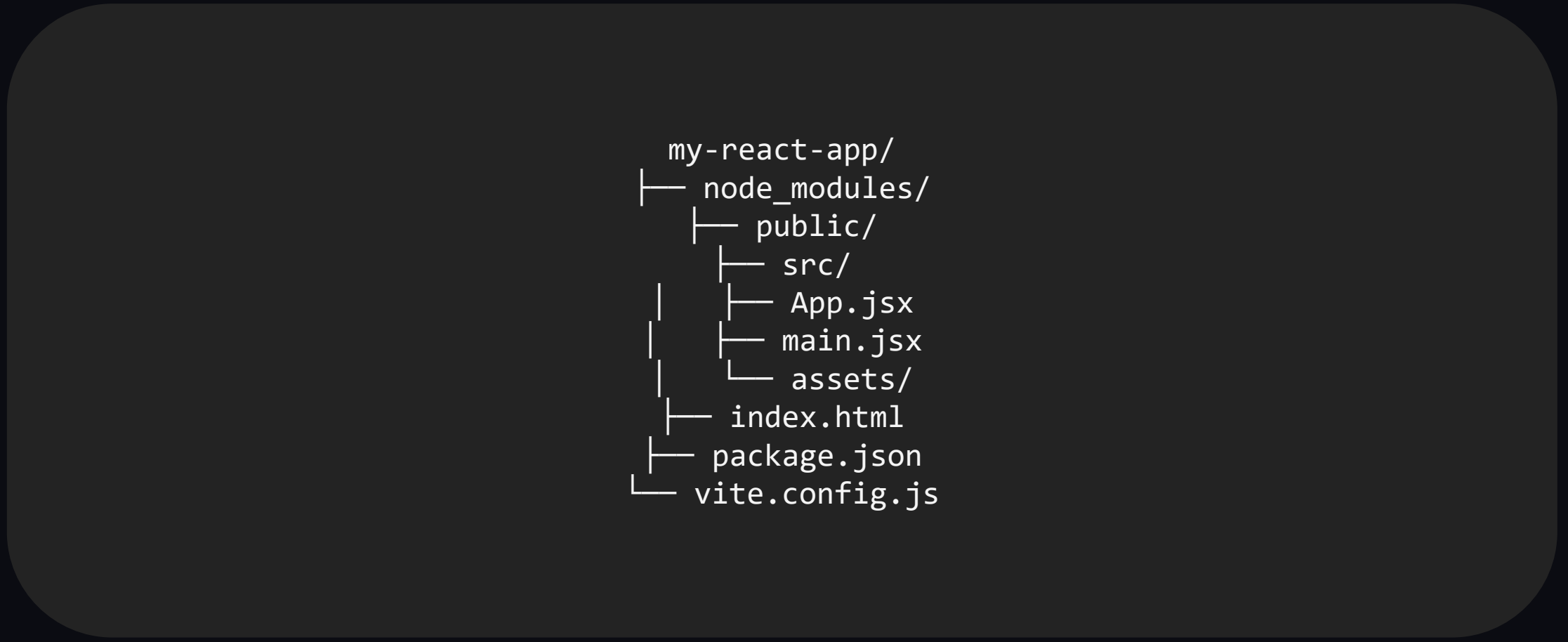
Start Vite's local development server

The terminal will show a local URL, usually http://localhost:5173

Open that URL in your browser

```
npm run dev
```

React Project Structure



Focus first on src/, App.jsx, and main.jsx.

Step 6 — Edit Your First React Component

Open src/App.jsx

Replace the existing component with a simple component:

Save the file and check the browser — Vite updates the page automatically.

```
function App() {
return <h1>Hello React!</h1>
}

export default App
```

Complete Installation Flow

```
node -v
npm -v
npm create vite@latest my-react-app
cd my-react-app
npm install
npm run dev
```

Browser → http://localhost:5173

You now have a working React development environment.

Common Installation Errors

‘node’ is not recognized → Node.js is not installed correctly or PATH needs refreshing

‘npm’ is not recognized → restart terminal / reinstall Node.js if needed

npm install fails → check internet, npm registry, and project folder

Port already in use → Vite can usually choose another available port

Permission errors → avoid running projects from restricted system folders

Quick Student Practice

1. Install Node.js LTS
2. Verify node -v and npm -v
3. Create a project named student-react-app
4. Run npm install
5. Run npm run dev
6. Change App.jsx to display your name and a welcome message
7. Stop the server with Ctrl + C

What's Next?

JSX — writing HTML-like syntax inside JavaScript

Components — building reusable UI pieces

Props — passing data between components

State — managing changing data

Events — handling user interactions

Hooks — useState, useEffect and more

Routing — building multi-page experiences